

Future work and limitation report:

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1. Introduction

1.1. Purpose of the Report

The purpose of this report is to provide a comprehensive analysis of the **CIRCEBot** project's current status, outlining both the **potential avenues for future work** and the **inherent limitations** of the system as implemented. The insights presented herein are intended to guide subsequent development efforts, inform resource allocation, and ensure the long-term viability and competitive advantage of the platform.

1.2. Scope

This report is scoped to cover the proposed **enhancements** and **research and development** areas for the CIRCEBot system, including hardware-software integration, AI/ML capabilities (object detection, path-planning), and UI/UX improvements. It will also

systematically detail the **key limitations** affecting the project, categorized as technical, resource-based, data-related, and regulatory/compliance constraints, which must be addressed for the system to achieve its full operational potential.

2. Future Work

2.1. Potential Enhancements

During the initial planning phase of this project, several advanced features were proposed but ultimately not implemented due to time limitations and development prioritization. One area of future improvement includes coordinated testing with the engineering team to validate full end-to-end communication between the physical Raspberry Pi onboard CIRCEBot and the backend control software. Another proposed enhancement is the integration of a lightweight AI model capable of performing automatic object detection on the satellite imagery used by the system, enabling more intelligent obstacle handling and terrain awareness. Additionally, the capability to stream CIRCEBot's onboard camera feed directly to the operator interface would significantly improve situational awareness and support real-time decision making during active operations.

2.2. Research and Development

To maintain system relevance and remain competitive with evolving industry standards, several areas warrant continued research and development. Advancements in autonomous navigation, including more sophisticated path-planning algorithms and dynamic obstacle detection, could significantly enhance CIRCEBot's operational reliability in real-world environments. Continued exploration into secure communication protocols, hardened embedded systems, and zero-trust architectures would strengthen the platform's cybersecurity posture as threats and adversarial techniques grow more complex. Additionally, incorporating modern machine learning techniques—such as real-time image classification, sensor fusion, and predictive analytics—may provide CIRCEBot with greater autonomy, improved situational awareness, and more adaptive decision-making capabilities. Ongoing refinement of UI/UX elements, data visualization tools, and operator feedback systems will also be essential to ensure that the platform remains user-friendly and effective for mission-critical applications.

2.3. Timeline for Future Work

A phased development timeline is recommended to guide the implementation of the proposed enhancements and ensure steady, manageable progress. In the **short term (0–3 months)**, efforts should focus on coordinated hardware–software integration testing with the engineering team and establishing secure, reliable communication between the physical Raspberry Pi and the backend system. During the **medium-term phase (3–6 months)**, development can shift

toward integrating lightweight AI-based object detection and improving the system's path-planning intelligence. In the **long term (6–12 months and beyond)**, work can be directed toward implementing real-time camera streaming, advancing autonomous decision-making capabilities, and refining the operator interface to support increasingly complex mission requirements.

3. Limitations

3.1. Technical Limitations

In its present state, the system references a static background containing a satellite image of the football field. The measurements of the background, and the ratio of grid-map to display, are therefore hard-coded and the image is inflexible. Therefore, the potential field of operations is limited to the TTU football field. In future iterations, adjustments may be made to allow dynamic scaling of the aerial image based on grid transformations. Future goals may include implementing AI image detection to automatically interpret relative distances from an aerial image. This would allow greater flexibility for choice of field of operations.

3.2. Resource Limitations

Incorporating a stream of aerial image inputs to display real-time view of the field of operations would be a trivial software task. However, testing and development of this feature is limited by drone access. To extend the project with AI image processing for automated terrain detection and obstacle avoidance, high resolution image inputs are required. Consistent access to a high resolution drone may prove prohibitively expensive.

In order to test the software within the whole Circe System, a physical CirceBot is required. The bot must be able to drag a substantial amount of ethernet cable and install it autonomously. The cable must support both the "repaired telecommunications" defined by the initial project requirements and simultaneously transfer data between software and hardware.

Budget is required to support hosting the browser interface on GCP for a consistent frontend development pipeline.

3.3. Data Limitations

The software is currently based on user inputs. To extend the project with AI image processing for automated terrain detection and obstacle avoidance, high resolution image inputs must be transferred across tcp. Any AI image processing used to automate pathfinding abilities will depend on high quality data inputs and consistent access to high performance computers.

Currently, the software is tested using a virtual simulation. However, in order to attach a physical robot to the network, rigorous testing is required so that data losses are detected. Introducing new hardware to the network may complicate data transfer on the client side.

3.4. Regulatory and Compliance Constraints

Future iterations of this project which depend upon flying drones will be subject to regulatory limitations.

4. Conclusion

4.1. Summary of Future Work

This report highlights a clear path forward for the CIRCEBot system, emphasizing necessary **short-term goals** like rigorous hardware-software integration testing and establishing secure communications. **Medium-term objectives** focus on introducing more sophisticated autonomy through lightweight AI-based object detection and improved path-planning intelligence. **Long-term goals** include implementing real-time camera streaming and advanced autonomous decision-making. These enhancements are critical for transforming CIRCEBot from a simulated prototype to a robust, mission-capable platform ready for diverse operational environments and evolving industry standards.

4.2. Implications of Limitations

The identified limitations, particularly the reliance on a **static operational field** (TTU football field) and constraints related to **drone access and cost** for high-resolution data, pose significant hurdles to achieving system flexibility and autonomy. The **technical inflexibility** of hard-coded grid-map ratios prevents easy deployment to new environments. **Resource limitations** (cost of high-resolution drones and physical testing bots) directly impede the implementation of AI-driven features like automated terrain detection. Furthermore, **data quality and access** are paramount; without consistent, high-quality data and computational resources, the system's ability to evolve its pathfinding and situational awareness through machine learning will be severely compromised. Addressing these limitations is paramount, as they currently confine the system to a proof-of-concept state rather than a universally deployable solution.

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